A Diagrammatic Language for Epistemic Geometry

Foundational Extensions, Invariants, and Certified Closure

Abstract

This document specifies a visual, diagrammatic language—*EGDL* (Epistemic Geometry Diagram Language)—designed to express structures appearing in Epistemic Geometry: metric interfaces between syntax and semantics, curvature as a representation gap, refinement dynamics (DRP), closure nets, finite-bank certification, coherent flows, and holonomy obstructions. The presentation is organized as *definitions* → *axioms* → *lemmas/theorems* → *invariants* → *structural predictions*. All constructions are compatible with LaTeX and TikZ, and the document includes a compact TikZ "micro-library" for publication-ready diagrams.

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1 Definitions

1.1 Core objects of Epistemic Geometry

Definition 1.1 (Metric interface)**.** A *metric interface* is a tuple

$$S = (L, O, X, e, j, \delta),$$

where:

- L is a (syntactic) language of candidates σ.
- O is an observation/target space (semantic outputs).
- X is a measurable/topological parameter space.
- $e : L \times X \to O$ is an evaluation map.
- $j : O \times X \to O$ is a reference/idealization operator (optional; sometimes implicit).
- δ is a metric on an ambient space containing O, or a family of pseudometrics.

Definition 1.2 (Error and curvatures)**.** Given a metric interface S, define the error of a candidate $\sigma \in L$ as

$$\mathrm{err}(\sigma) := \sup_{x \in X} \delta\big(e(\sigma, x),\, j(o^\star, x)\big),$$

where $o^\star$ denotes an ideal semantic object (or a specified target family). The *global curvature* of S is

$$\kappa_S := \inf_{\sigma \in L} \mathrm{err}(\sigma).$$

Resource-bounded variants arise from filtrations $L_{\leq r} \subseteq L$, giving

$$\kappa_{S,r} := \inf_{\sigma \in L_{\leq r}} \mathrm{err}(\sigma).$$

Remark 1.1. The particular choice of sup over X and the presence of j may vary across instances (worst-case vs windowed, ideal vs empirical, etc.). The diagram language in section 1.4 treats these choices as region and label data.

Definition 1.3 (Windows)**.** A *window* is a specified subobject $K \subseteq X$ (often compact or measurable). The *window-restricted error* is

$$\mathrm{err}_K(\sigma) := \sup_{x \in K} \delta\big(e(\sigma, x),\, j(o^\star, x)\big).$$

Definition 1.4 (cGCNF open families)**.** A *cGCNF predicate* is represented by finite Boolean combinations of basic opens obtained as preimages of opens along continuous maps. Equivalently, it defines a subobject that is stable under pullback and finite limits/colimits available in the ambient setting. The intended semantics is an "open" (in the topological sense) or a frame element (in a point-free setting).

Definition 1.5 (Derivational Refinement Principle (DRP))**.** A *refinement operator* is a map $T : L \to L$ satisfying:

(i) (*Derivability preservation*) If σ is derivable/certified, then $T(\sigma)$ is derivable/certified in the same system.

(ii) (*Non-expansive error*) $\mathrm{err}(T(\sigma)) \leq \mathrm{err}(\sigma)$.

(iii) (*Orbit convergence in infimum*) For each σ, the infimum of err along the orbit $\{T^n(\sigma)\}_{n \geq 0}$ equals the best achievable error along that orbit.

1.2 Closure nets, atlases, protocols, finite banks

Definition 1.6 (Closure net)**.** A *closure net* is a typed directed graph (or higher graph) whose nodes are modules (e.g. a kernel of certification, a geometric normal form layer, a bank of templates, atlas and protocol components) and whose edges are typed transformations between modules. The net supports:

- local certification and local refinement,
- transfer across finite banks (via moduli),
- atlas gluing data (groupoid/cocycle constraints),
- protocol dependence (state/history).

Definition 1.7 (Atlas and atlas holonomy)**.** An *atlas* on a domain is a cover $\{U_i\}$ equipped with local charts and transition data g_{ij} on overlaps $U_i \cap U_j$. On triple overlaps $U_i \cap U_j \cap U_k$, define a cocycle

$$h_{ijk} \coloneqq g_{ij}\, g_{jk}\, g_{ki}.$$

The corresponding *atlas holonomy* is the gauge-invariant class determined by the family $\{h_{ijk}\}$.

Definition 1.8 (Protocol holonomy)**.** A *protocol* is a stateful (history-dependent) evaluation mechanism. *Protocol holonomy* is the dependence of the output on nontrivial loops in the space of states/histories, up to program-level equivalence (as opposed to index-level relabeling).

Definition 1.9 (Finite-bank certification)**.** A *finite bank* is a compact template space $\mathcal{T}$ equipped with an ε-net $B_\varepsilon \subseteq \mathcal{T}$. A *transfer modulus* $\omega(\varepsilon)$ bounds the semantic discrepancy between ideal certification over $\mathcal{T}$ and certification restricted to the finite bank B_ε. The induced discrepancy region is referred to as a *gray zone*.

1.3 Foundational extensions (pure mathematics / foundations)

Definition 1.10 (Explicit metatheory parameter)**.** Fix a metatheory label $\mathfrak{F} \in \{\mathrm{ZFC}, \mathrm{ETCS}, \mathrm{MK}, \mathrm{MLTT}, \mathrm{HoTT}, \ldots\}$ A proof judgment is written $\vdash_{\mathfrak{F}} \varphi$. In the present setting, any statement tagged *[Proved]* is treated as $\vdash_{\mathfrak{F}} \varphi$ with $\mathfrak{F}$ declared.

Definition 1.11 (Geometric semantics via locales/toposes)**.** A cGCNF predicate is interpreted as an element of a frame of opens in a locale $X \in \mathbf{Loc}$, or as a subobject in a topos of sheaves $\mathbf{Sh}(X)$. Pullbacks along parameter maps implement substitution of variables/parameters.

Definition 1.12 (Window site and modal operator)**.** Let $\mathcal{K}$ be a category of windows (objects are windows, morphisms are refinements/embeddings). A *local modality* $\Box_K$ is assigned to each $K \in \mathrm{Ob}(\mathcal{K})$, with intended reading:

$$\Box_K(\varphi) \ : \ \text{“}\varphi \text{ holds (is certifiable) when restricted to window } K\text{”}.$$

Definition 1.13 (A 2-category of interfaces)**.** Define a (strict or weak) 2-category **EG** where:

- objects are metric interfaces S,
- 1-morphisms are non-expansive structure-preserving transformations between interfaces (e.g. Lipschitz maps transporting evaluation and reference data),

- 2-morphisms are gauge transformations (changes of trivialization, conjugacies of transitions, protocol equivalences).

Definition 1.14 (Curvature spectrum)**.** Given a target class C (of semantic objects or constraints), define the resource-indexed curvature function

$$f_C(r) \coloneqq \kappa^{\sup}_{S,r}(C),$$

viewed as a complexity-like invariant of C relative to S.

1.4 EGDL: a diagrammatic language

Definition 1.15 (EGDL diagrams)**.** An *EGDL diagram* is a planar embedded directed graph equipped with:

(i) an ordered boundary of input/output ports,

(ii) typed wires (objects) and typed boxes/vertices (morphisms),

(iii) optional 2-cells between parallel morphisms,

(iv) optional regions (windows, gray zones, geometric opens, constraint leaves),

(v) numeric or symbolic decorations (e.g. $r, \varepsilon, \kappa, \omega(\varepsilon), V(K), h_{ijk}$).

Definition 1.16 (Type alphabet)**.** The minimal type alphabet used below:

I (interface), P (predicate/open), K (window), R (resource), B (bank), A (atlas), $\mathsf{\Pi}$ (protocol).

Definition 1.17 (Compositions)**.** EGDL supports three compositions:

(i) sequential composition $D_2 \circ D_1$ by gluing outputs to inputs,

(ii) monoidal composition $D_1 \otimes D_2$ by drawing in parallel,

(iii) region gluing (atlas/cover): identify region boundaries representing overlaps $U_i \cap U_j$ to encode transition data and holonomy.

Remark 1.2. The intended algebraic completion is a free (symmetric) monoidal 2-category generated by a signature of typed generators plus region modalities. The evaluation of a diagram is a functor from this free structure into the target 2-category **EG**.

1.5 EGDL diagrams (TikZ-ready figures)

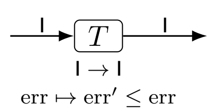

Figure 1: A DRP refinement step T as a typed string-diagram box.

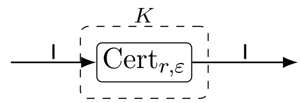

Figure 2: Window restriction via a region K containing a resource-bounded certification operator.

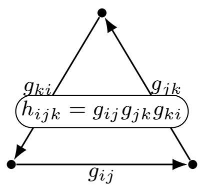

Figure 3: Atlas holonomy on a triple overlap represented by a cocycle label h_{ijk}.

2 Axioms

2.1 Core closure / certification axioms

Axiom 2.1 (Soundness-only certification)**.** A certification operator $\mathrm{Cert}_{r,\varepsilon}$ is sound: whenever it outputs "certified," the certified statement holds within its declared tolerance and window. Completeness is not assumed.

Axiom 2.2 (Typed refinement)**.** All refinement operators T are typed as endomorphisms $T : \mathsf{I} \to \mathsf{I}$ (or suitable endomorphisms of the underlying syntactic component) and preserve derivability constraints.

Axiom 2.3 (Bank transfer modulus)**.** Finite-bank transfer is governed by a modulus $\omega(\varepsilon)$ such that for each ideal certificate over a compact template space there exists a bank-based certificate with a discrepancy bounded by $\omega(\varepsilon)$.

2.2 Geometric and foundational axioms

Axiom 2.4 (Explicit metatheory)**.** Any proof judgment is recorded as $\vdash_{\mathfrak{F}} \varphi$ with an explicit metatheory parameter $\mathfrak{F}$.

Axiom 2.5 (Geometric semantics)**.** cGCNF predicates are interpreted as opens in a locale or as subobjects in a sheaf topos, and substitution acts by pullback.

Axiom 2.6 (Holonomy as a gauge invariant)**.** Atlas and protocol holonomies are invariants under gauge 2-morphisms in **EG**.

2.3 EGDL axioms (diagrammatic calculus)

Axiom 2.7 (Planar isotopy invariance)**.** The value D of an EGDL diagram depends only on its planar isotopy class, with boundary port order fixed.

Axiom 2.8 (2-cells implement weak equality)**.** Given a 2-cell $\alpha : f \Rightarrow g$, contexts invariant under gauge treat f and g as equivalent.

Axiom 2.9 (Regions implement locality)**.** A region labeled K restricts all enclosed semantics to window K; nested regions compose by intersection.

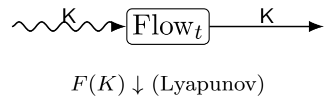

Figure 4: A "Feynman-like" coherent-flow step: a map Flow_t propagating a window/state K while decreasing a free-energy functional F.

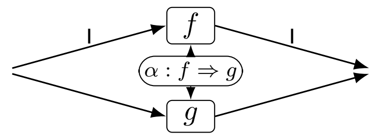

Figure 5: A 2-cell (gauge) between parallel 1-morphisms in **EG**.

3 Lemmas and Theorems

3.1 Diagrammatic rewrite rules

Lemma 3.1 (Window absorption). *For window regions K, K',*

$$\mathrm{Res}_K \circ \mathrm{Res}_{K'} \cong \mathrm{Res}_{K \cap K'}.$$

In EGDL: nested or successive window frames merge into a single frame labeled $K \cap K'$.

Lemma 3.2 (DRP monotonicity). *If T satisfies DRP, then* $\mathrm{err}(T(\sigma)) \leq \mathrm{err}(\sigma)$ *for all σ, and hence κ_S is a lower bound for all refinement orbits.*

Lemma 3.3 (Gauge conjugacy for atlas cocycles). *Under chart gauge changes g_i, cocycle labels satisfy*

$$h_{ijk} \sim g_i \, h_{ijk} \, g_i^{-1}.$$

The holonomy class is invariant under this relation.

3.2 Pillar theorem schemas present in the corpus (abstract form)

Theorem 3.1 (Diagonal obstruction schema). *In any setting admitting an arithmetization of a "zero-error provability" predicate* $\mathrm{Prov}_{0,r}$ *satisfying soundness and basic representability, there exists a sentence G such that G holds yet* $\neg\mathrm{Prov}_{0,r}(\ulcorner G \urcorner)$*. Consequently, for the corresponding arithmetical target class C_{arith},*

$$\kappa^{\sup}_{S,r}(C_{\mathrm{arith}}) > 0.$$

Theorem 3.2 (Fragmentation barrier schema). *There exist families of semantic sets whose exact compilation into a fixed normal form requires an exponential number of components (e.g. 2^n pieces for unions of intervals with 2^n connected components). In EGDL this manifests as a lower bound on the number of disjoint subdiagrams required for an exact* Comp *representation.*

Theorem 3.3 (Coherent-flow Lyapunov schema). *Let $F(K) = \alpha E_{\mathrm{ctr}}(K) + \beta|K| - \gamma \log V(K)$ be a free-energy functional over windows/states. For a coherent-flow operator U (or continuous dynamics* Flow_t*), the value of F decreases along iterations/trajectories, and fixed points correspond to coherent "islands" (local minima).*

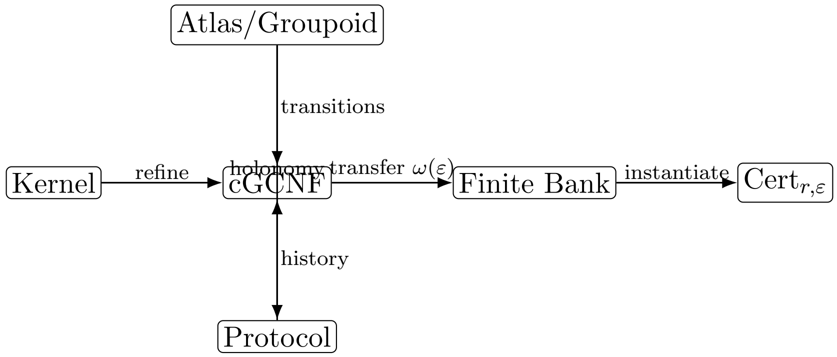

Figure 6: A schematic closure net: modules and typed edges linking normal forms, finite banks, certification, atlases (holonomy), and protocol dependence.

Theorem 3.4 (KL-curvature schema)**.** *Given a statistical manifold $\mathcal{M}$ (e.g. a product family encoding an independence constraint) and a distribution p, define*

$$\kappa_{\mathcal{M}}(p) := \inf_{q \in \mathcal{M}} D_{\mathrm{KL}}(p \| q).$$

On constraint leaves defined by fixed marginals, the KL-based flow has a Lyapunov identity and a unique equilibrium (the information projection).

Theorem 3.5 (Finite-bank transfer schema)**.** *Let $\mathcal{T}$ be compact and $B_\varepsilon \subseteq \mathcal{T}$ an ε-net. If an existence/certification statement holds uniformly over $\mathcal{T}$, then a corresponding statement holds over B_ε with slack bounded by $\omega(\varepsilon)$.*

3.3 Proposed foundational theorems

Theorem 3.6 (Representation via geometric semantics)**.** *For a suitable parameter object X, the fragment of predicates generated by cGCNF constructors is represented by the sub-frame of $\mathcal{O}(X)$ generated by basic cylinder opens and closed under finite limits/colimits available in the chosen semantic category (topological, locale, or sheaf-theoretic).*

Theorem 3.7 (Holonomy as a cohomology class)**.** *Given a cover $\{U_i\}$ and allowed transition automorphisms forming a sheaf/group object $\mathcal{G}$, the atlas holonomy defines a (possibly non-abelian) Čech class*

$$[h] \in \check{H}^2(\{U_i\}; \mathcal{G}),$$

invariant under gauge. Nontriviality of $[h]$ obstructs global trivialization.

Theorem 3.8 (Curvature spectrum as an equivalence invariant)**.** *Let $F : \mathbf{EG} \to \mathbf{EG}$ be an equivalence (weak 2-equivalence) preserving resource filtrations. Then for each target class C,*

$$f_C(r) \simeq f_{F(C)}(r)$$

up to the equivalence notion induced by F. Hence the curvature spectrum is an invariant of the equivalence class of the interface.

4 Invariants

Definition 4.1 (Holonomy invariants).

- *Atlas holonomy*: gauge class of cocycles h_{ijk}.
- *Protocol holonomy*: gauge class of history-dependent loops in protocol space.

Definition 4.2 (Curvature invariants).

- global curvature κ_S,
- resource-bounded curvature $\kappa_{S,r}$,
- worst-case/target-class curvature $\kappa_{S,r}^{\sup}(C)$,
- curvature spectrum $f_C(r)$.

Definition 4.3 (Finite-bank invariants).

- covering radius ε (or ε_N as a function of bank size),
- transfer modulus $\omega(\varepsilon)$,
- measure/volume of the induced gray zone.

Definition 4.4 (Diagrammatic invariants). Given an EGDL diagram D:

- *diagram holonomy*: product of transition labels around marked loops,
- *diagram curvature*: infimum of error labels reachable by DRP rewrites,
- *structural complexity*: size, width, treewidth, crossing number (if nonplanar extensions are enabled).

5 Structural predictions

Proposition 5.1 (Cycles as nontrivial globality witnesses). *In the absence of loops (in a DAG-like fragment) and with trivial cocycle data, EGDL diagrams tend to admit global trivializations of local data. Nontrivial holonomy is represented by an unavoidable loop label that survives gauge rewrites.*

Proposition 5.2 (Fragmentation implies subdiagram proliferation). *Exact compilation of highly disconnected semantic sets enforces a lower bound on the number of disjoint subdiagrams appearing in any EGDL representation of* Comp, *mirroring combinatorial fragmentation barriers.*

Proposition 5.3 (Finite-bank quantization signature). *Introducing a finite bank inserts a* $\mathrm{Bank}_\varepsilon$ *stage and an explicit gray-zone region whose scale is controlled by* $\omega(\varepsilon)$*. As* $\varepsilon \to 0$*, the gray zone contracts according to the modulus.*

Proposition 5.4 (Diagonal obstruction produces curvature floors). *Whenever internal diagonalization applies, the spectrum* $f_{C_{\mathrm{arith}}}(r)$ *exhibits a positive floor (possibly resource-dependent). In EGDL, this appears as a minimal error label invariant under all DRP rewrites within the declared resources.*

Appendix A: EGDL TikZ micro-library (copy/paste)

```
\usepackage{tikz}
\usetikzlibrary{arrows.meta,calc,decorations.pathmorphing,positioning,fit,backgrounds,cd}
\usepackage{tikz-cd}

\tikzset{
  egwire/.style={-Latex, line width=0.6pt},
  egtype/.style={font=\scriptsize, inner sep=1pt},
  egbox/.style={draw, rounded corners=2pt, minimum height=12pt, minimum width=18pt, inner
      ↪  sep=2pt},
  egdot/.style={circle, fill, inner sep=1.2pt},
  egregion/.style={draw, dashed, rounded corners=3pt, inner sep=6pt},
  eggray/.style={draw, rounded corners=3pt, fill=black!10, inner sep=6pt},
  eg2cell/.style={draw, rounded corners=6pt, fill=white, inner sep=2pt, font=\scriptsize},
      ↪
  egsnake/.style={decorate,decoration={snake,amplitude=0.6mm,segment length=3mm}},
}
```

Appendix B: Pair revision as operators (Cantor, Turing, Gödel, Boltzmann, Riveros)

Definition 5.1 (Review operators)**.** Let $\mathcal{T}$ denote the theory object (definitions, axioms, rewrite rules, invariants). Define five "review operators"

$$\mathsf{C}, \mathsf{T}, \mathsf{G}, \mathsf{B}, \mathsf{R} : \mathcal{T} \to \mathcal{D}$$

into a diagnostic space $\mathcal{D}$, interpreted as follows:

- C (Cantor-profile): extracts cardinal/cover/locale strata and invariants of size.
- T (Turing-profile): extracts resource filtrations and the curvature spectrum $f_C(r)$.
- G (Gödel-profile): extracts arithmetization points and diagonal obstructions.
- B (Boltzmann-profile): extracts entropy/free-energy structure and Lyapunov monotonicity.
- R (Riveros-profile): extracts closure-net typing, auditability, and holonomy obstructions.

Proposition 5.5 (Composite verdict pattern)**.** *The composite diagnostic*

$$\mathsf{V} := \mathsf{C} \otimes \mathsf{T} \otimes \mathsf{G} \otimes \mathsf{B} \otimes \mathsf{R}$$

stabilizes on three persistent signatures:

(i) (Size/cover signature) a stratification by windows, covers, and bank radii ε,

(ii) (Resource signature) an organizing spectrum $r \mapsto f_C(r)$,

(iii) (Obstruction signature) holonomy/diagonalization as nontrivial globality witnesses.

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[6] O. Riveros, *Continuous Epistemic Geometry* (manuscript).